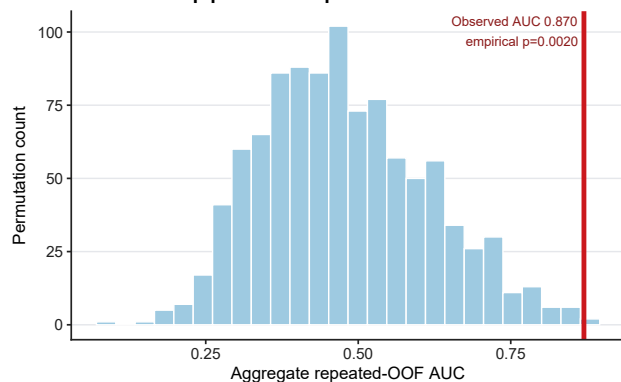
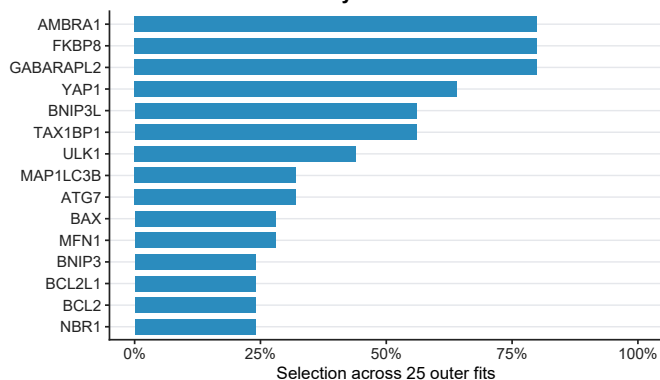
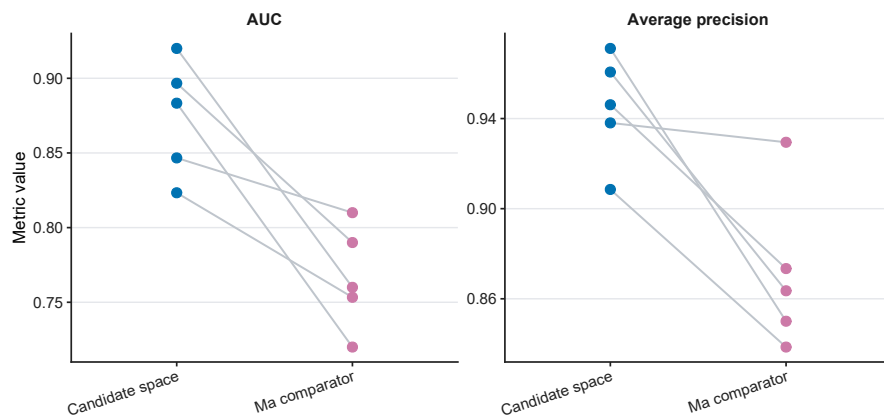


A**Matched full-pipeline label-permutation null****B****Feature-selection stability****C****Paired repeat-level performance**

Identical outer folds were used for both models

**D****Aggregated cross-fitted ROC**

Paired fixed-prediction comparison

— Candidate space · AUC 0.870

— Ma comparator · AUC 0.813

